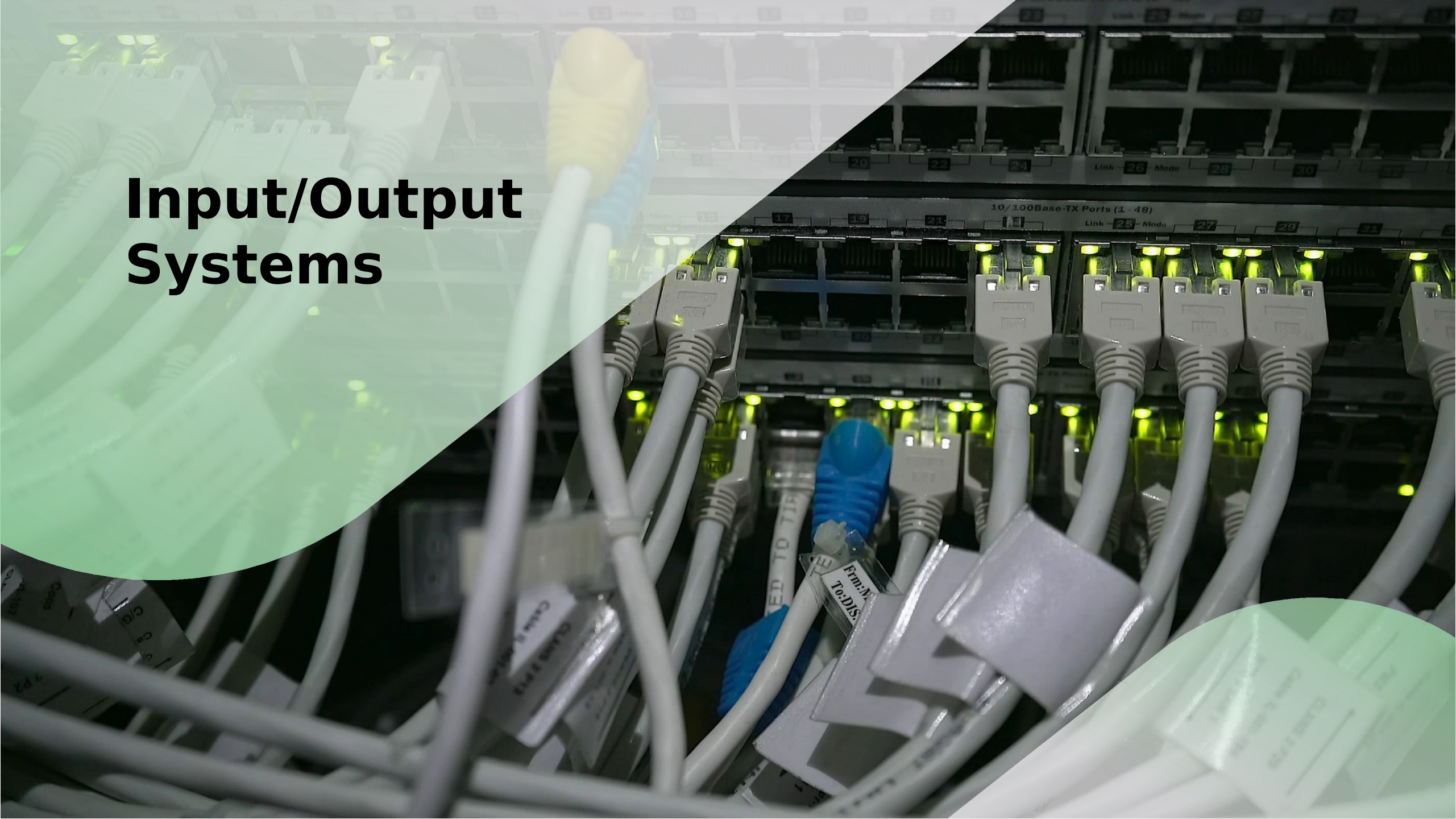


Input/Output Systems



Objectives

- Understand how I/O systems work, including I/O methods and architectures.
- Become familiar with storage media, and the differences in their respective formats.
- Understand how RAID improves disk performance and reliability, and which RAID systems are most useful today.
- Be familiar with emerging data storage technologies and the barriers that remain to be overcome.

Amdahl's Law

- The overall performance of a system is a result of the interaction of all of its components.
- System performance is most effectively improved when the performance of the most heavily used components is improved.
- This idea is quantified by Amdahl's Law:

$$S = \frac{1}{(1 - f) + (f/k)}$$

where S is the overall speedup;
 f is the fraction of work performed
by a faster component; and k is the
speedup of the faster component.

Amdahl's Law

- Amdahl's Law gives us a handy way to estimate the performance improvement we can expect when we upgrade a system component.
- On a large system, suppose we can upgrade a CPU to make it 50% faster for \$10,000 or upgrade its disk drives for \$7,000 to make them 150% faster.
- Processes spend 70% of their time running in the CPU and 30% of their time waiting for disk service.
- An upgrade of which component would offer the greater benefit for the lesser cost?

Amdahl's Law

- The processor option offers a 30% speedup:

$$f = 0.70, k = 1.5, \text{ so } S = \frac{1}{(1 - 0.7) + (0.7/1.5)} = 1.30$$

- And the disk drive option gives a 22% speedup:

$$f = 0.30, k = 2.5, \text{ so } S = \frac{1}{(1 - 0.3) + (0.3/2.5)} \approx 1.22$$

- Each 1% of improvement for the processor costs \$333, and for the disk a 1% improvement costs \$318.

I/O ARCHITECTURES

- Blocks of main memory that are devoted to I/O functions
- Buses that provide the means of moving data into and out of the system
- Control modules in the host and in peripheral devices
- Interfaces to external components such as keyboards and disks
- Cabling or communications links between the host system and its peripherals

I/O Control Methods

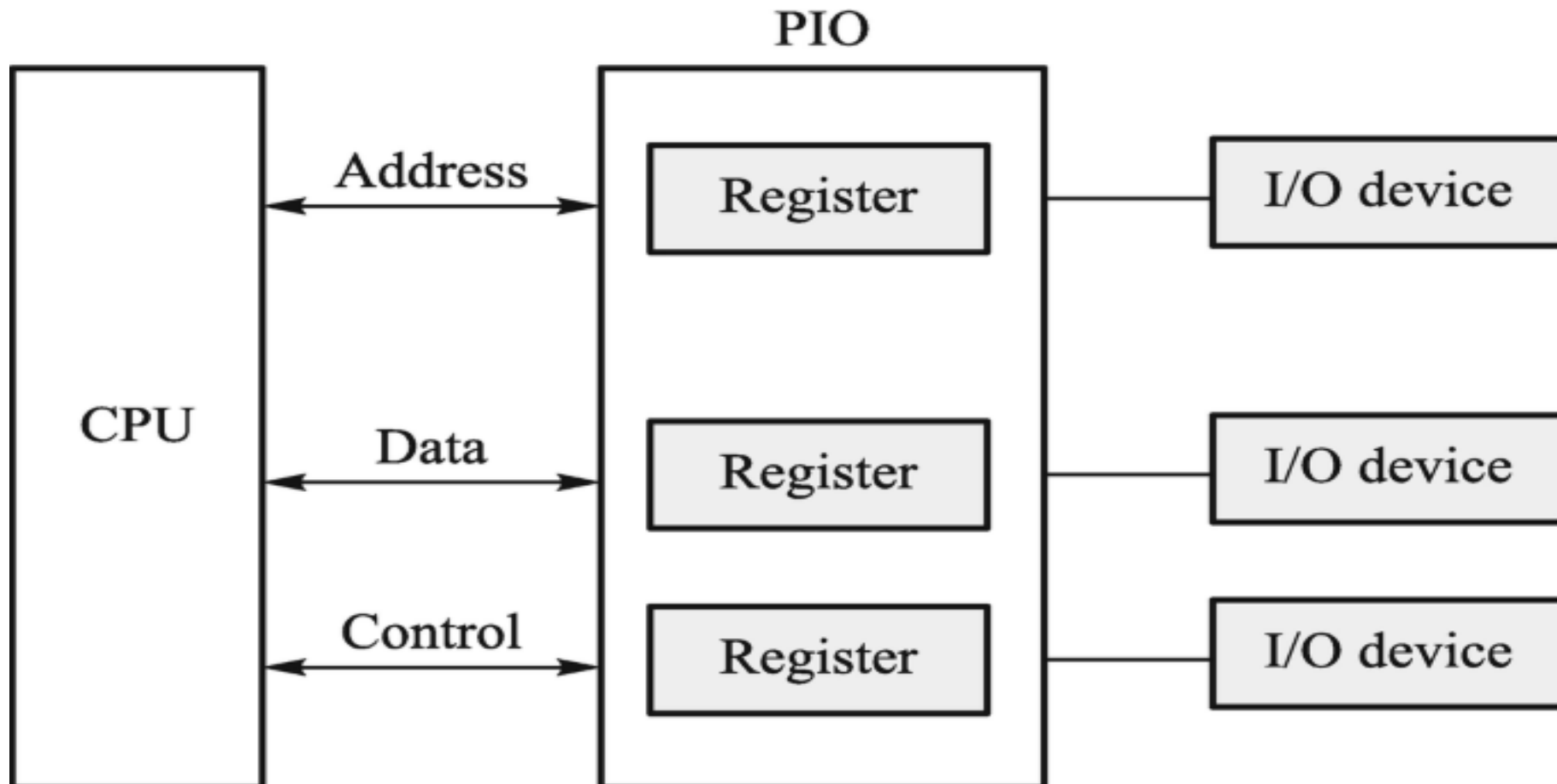
1. Programmed I/O
2. Interrupt-driven I/O
3. Memory-mapped I/O
4. Direct memory access
5. Channel-attached I/O

I/O Architectures

- I/O can be controlled in five general ways
 - 1. Programmed I/O** reserves a register for each I/O device. Each register is continually polled to detect data arrival
 - 2. Interrupt-Driven I/O** allows the CPU to do other things until I/O is requested
 - 3. Memory-Mapped I/O** shares memory address space between I/O devices and program memory
 - 4. Direct Memory Access (DMA)** offloads I/O processing to a special-purpose chip that takes care of the details
 - 5. Channel I/O** uses dedicated I/O processors

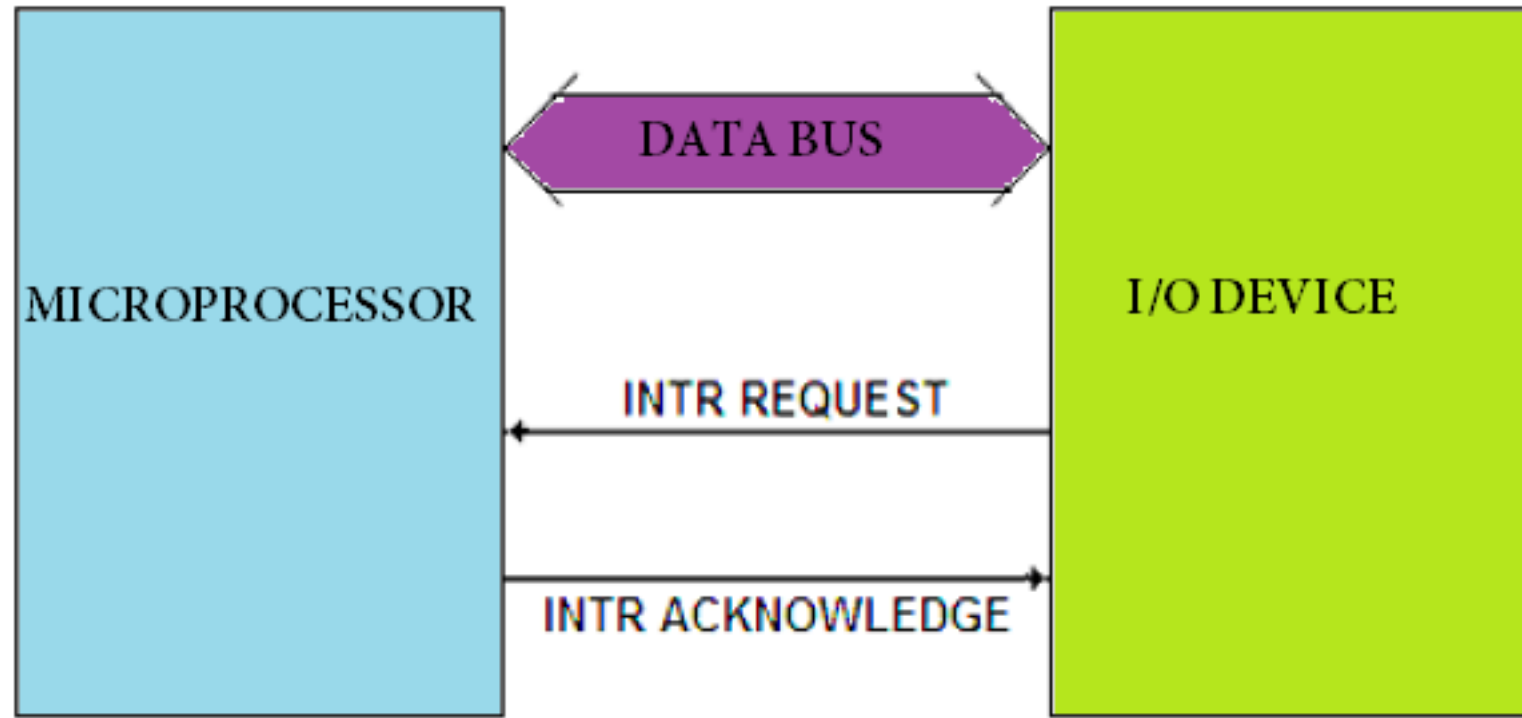
Programmed I/O

Programmed I/O reserves a register for each I/O device. Each register is continually polled to detect data arrival



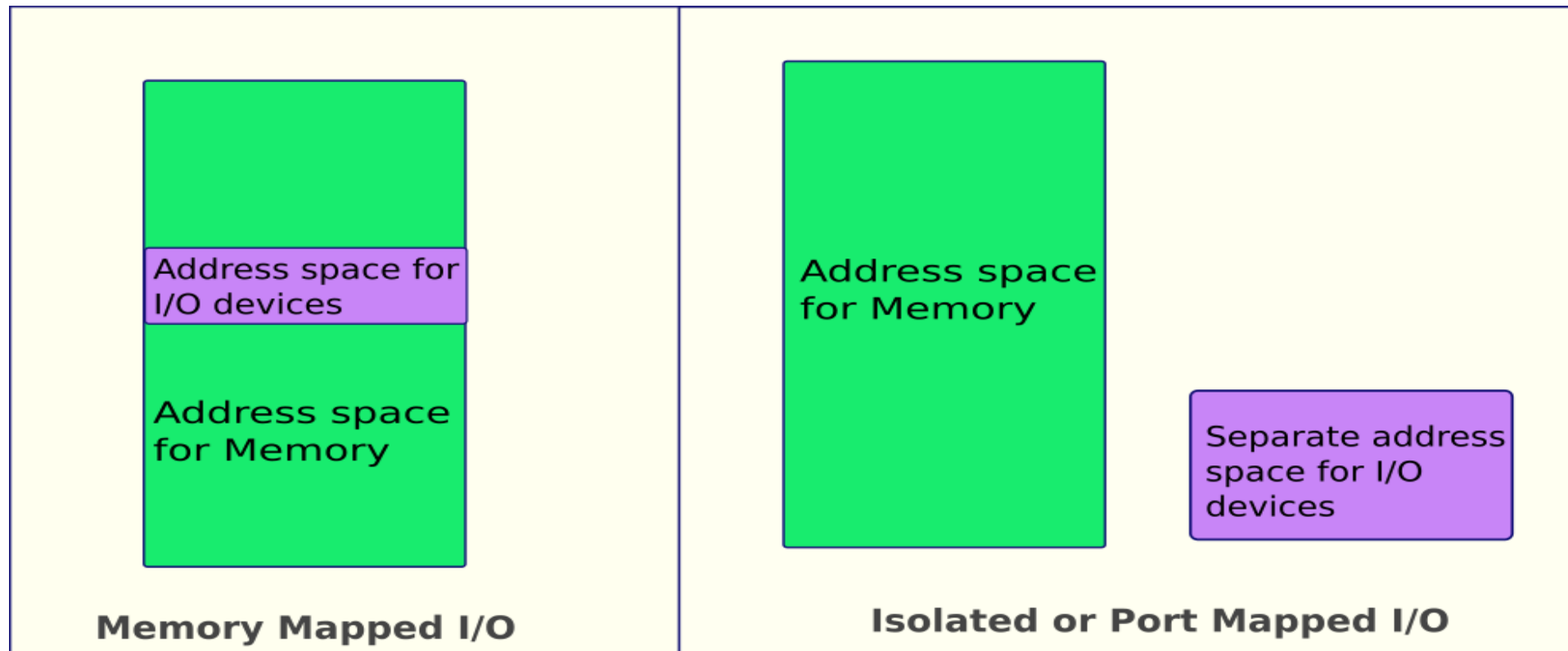
Interrupt-Driven I/O

***Interrupt-Driven I/O* allows the CPU to do other things until I/O is requested**



Memory-Mapped I/O

Memory-Mapped I/O shares memory address space between I/O devices and program memory.

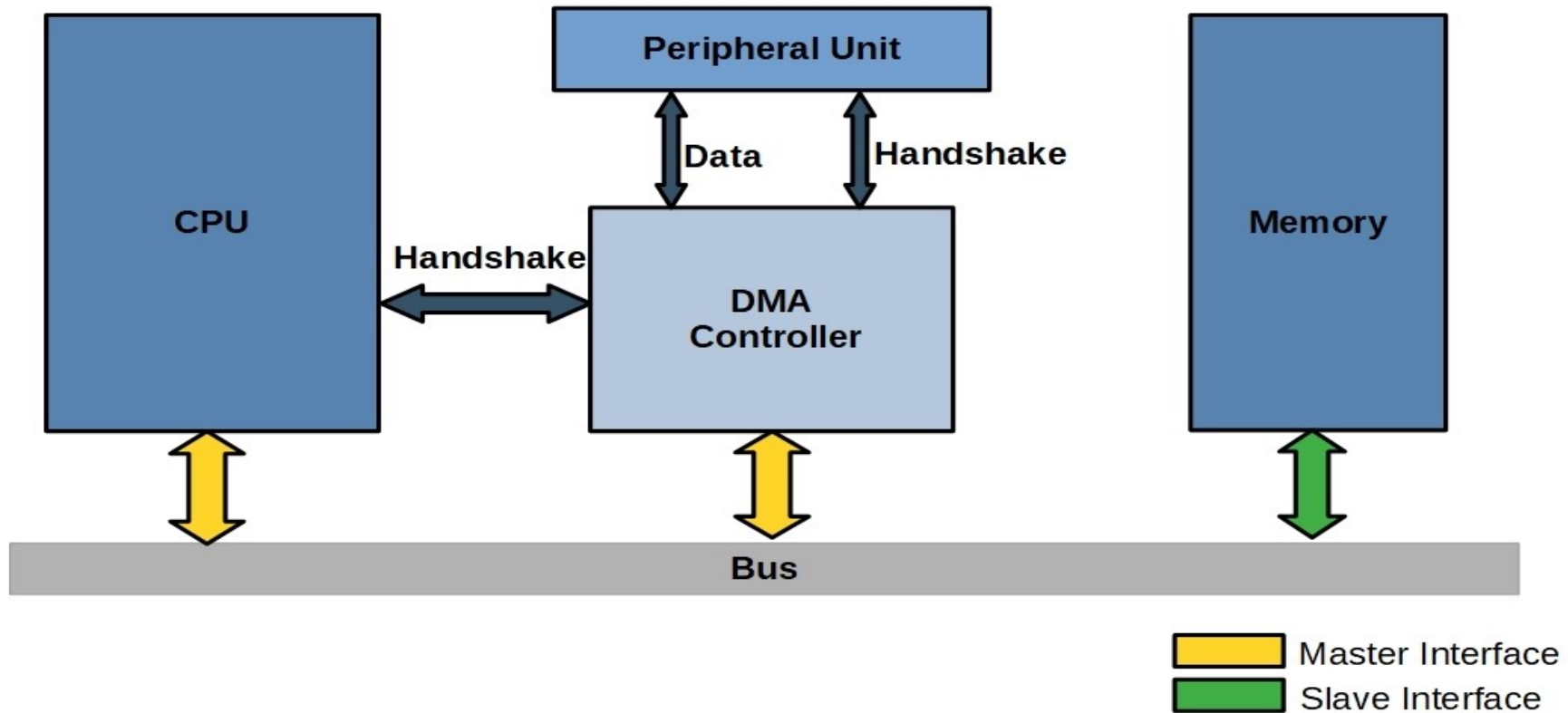


Memory-Mapped I/O

- In memory-mapped I/O devices and main memory share the same address space
 - Each I/O device has its own reserved block of memory
 - Memory-mapped I/O therefore looks just like a memory access from the point of view of the CPU
 - Thus the same instructions to move data to and from both I/O and memory, greatly simplifying system design
- In small systems the low-level details of the data transfers are offloaded to the I/O controllers built into the I/O devices.

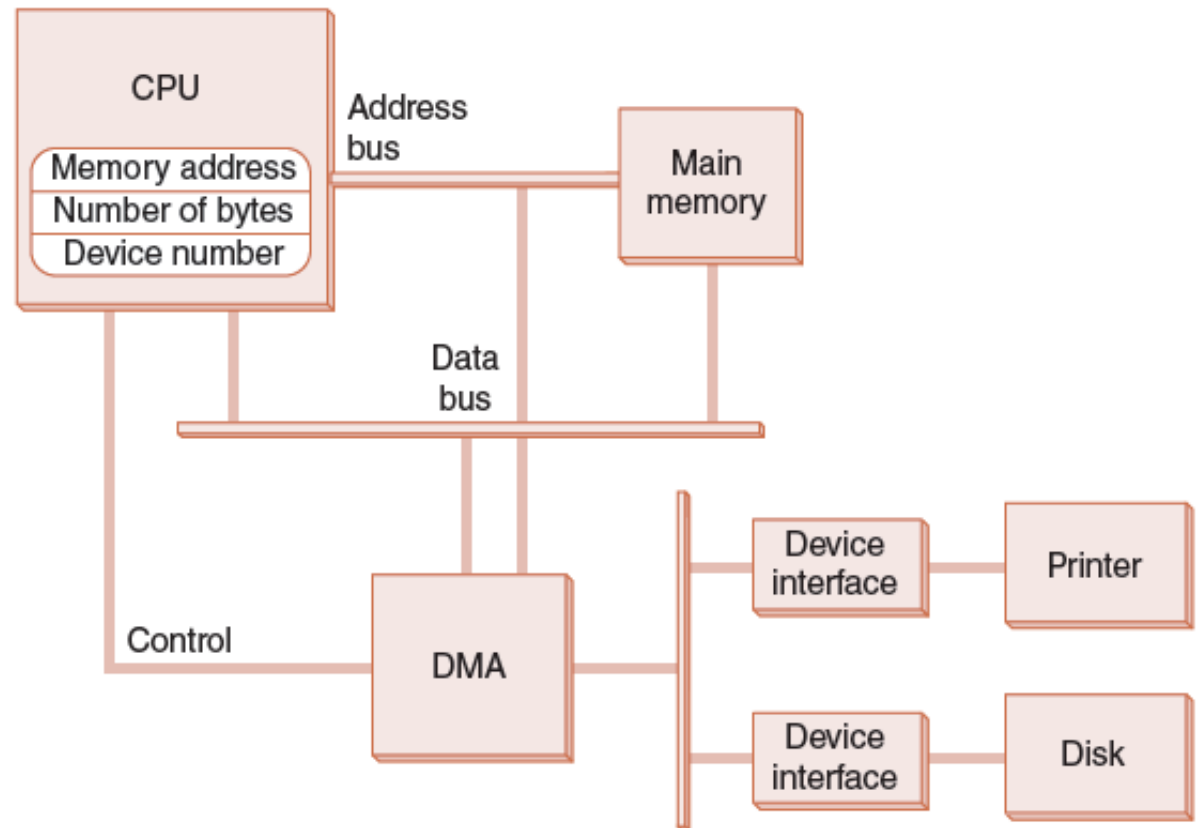
Direct Memory Access (DMA)

Direct Memory Access (DMA) offloads I/O processing to a special-purpose chip that takes care of the details. Channel I/O uses dedicated I/O processors

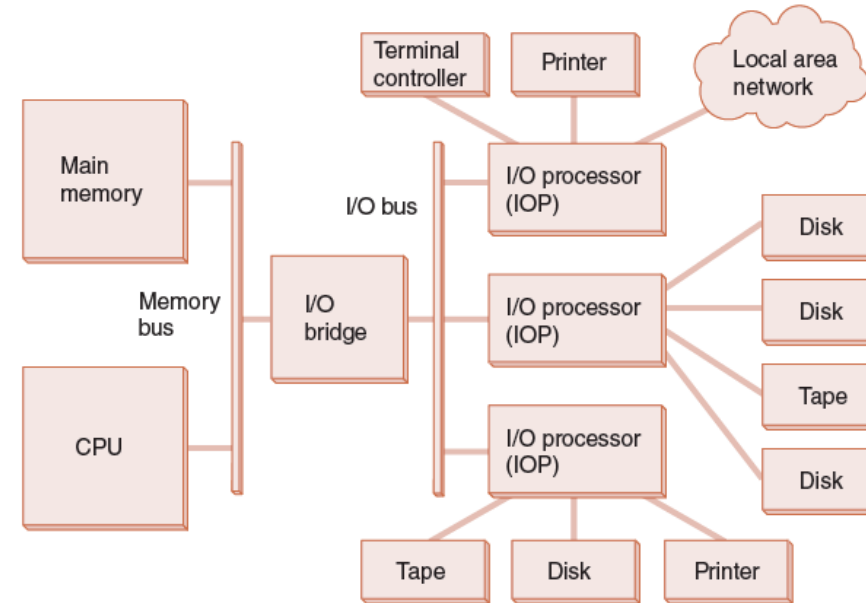
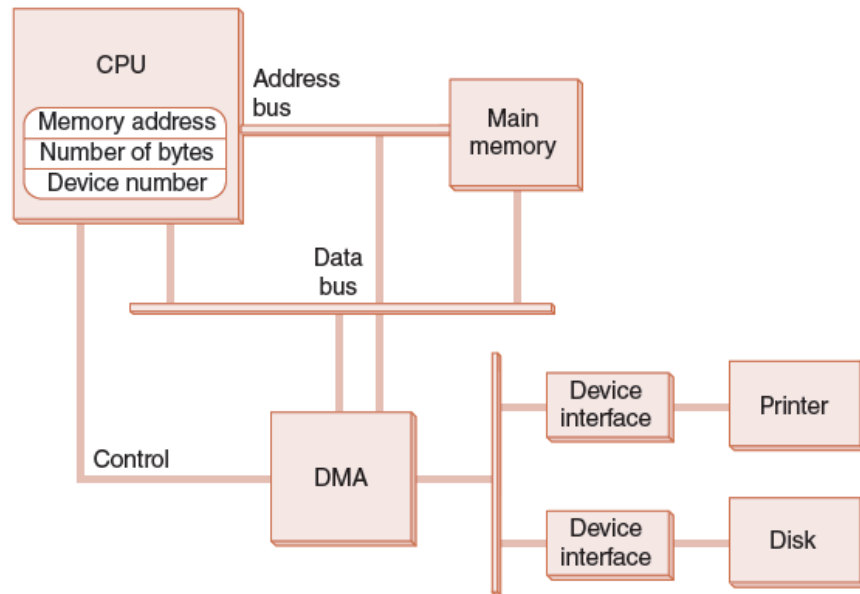


Direct Memory Access (DMA)

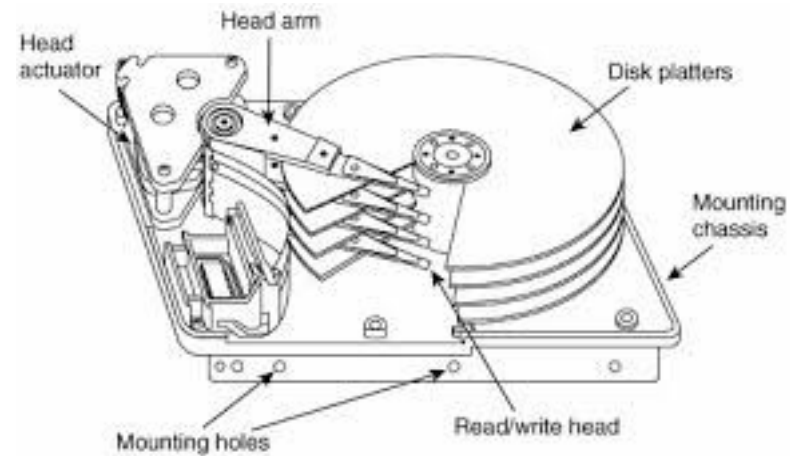
- This is a DMA configuration.
- Notice that the DMA and the CPU share the bus.
- The DMA runs at a higher priority and steals memory cycles from the CPU.



DMA vs CHANNEL I/O CONFIGURATION

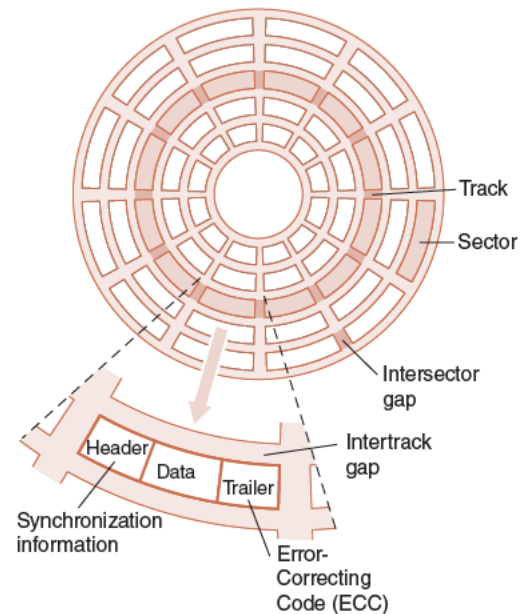


Rigid Disk Drive



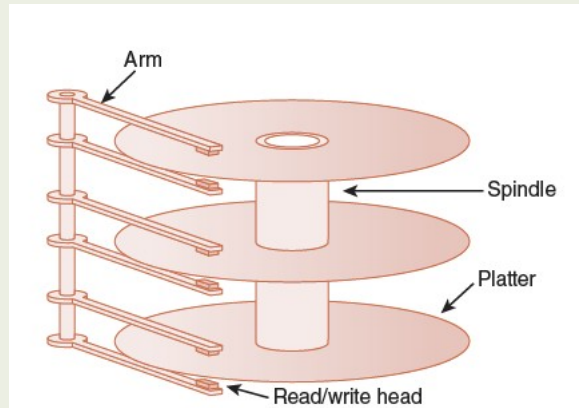
Rigid Disk Drives

- Disk tracks are numbered from the outside edge, starting with zero.

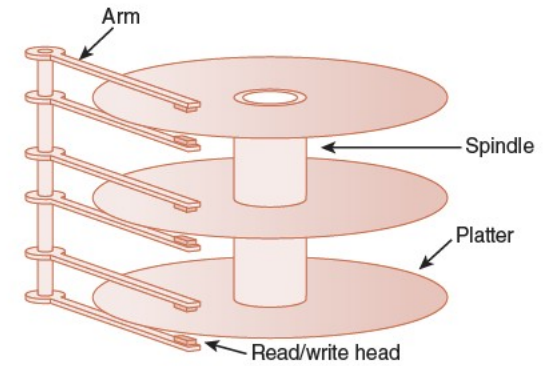


Rigid Disk Drives

- Hard disk platters are mounted on spindles.
- Read/write heads are mounted on a comb that swings radially to read the disk.



Rigid Disk Drives



- There are a number of electromechanical properties of hard disk drives that determine how fast its data can be accessed.
- *Seek time* is the time that it takes for a disk arm to move into position over the desired cylinder.
- *Rotational delay* is the time that it takes for the desired sector to move into position beneath the read/write head.
- $\text{Seek time} + \text{rotational delay} = \text{access time}$.

Solid State Drives



Solid State Drives

- SSD access time and transfer rates are *typically* 100 times faster than magnetic disk, but slower than onboard RAM by a factor of 100,000.
 - These numbers vary widely among manufacturers and interface methods.
- Unlike RAM, flash is block-addressable (like disk drives).
 - The duty cycle of flash is between 30,000 and 1,000,000 updates to a block.
 - Updates are spread over the entire medium through *wear leveling* to prolong the life of the SSD.

Magnetic Tape

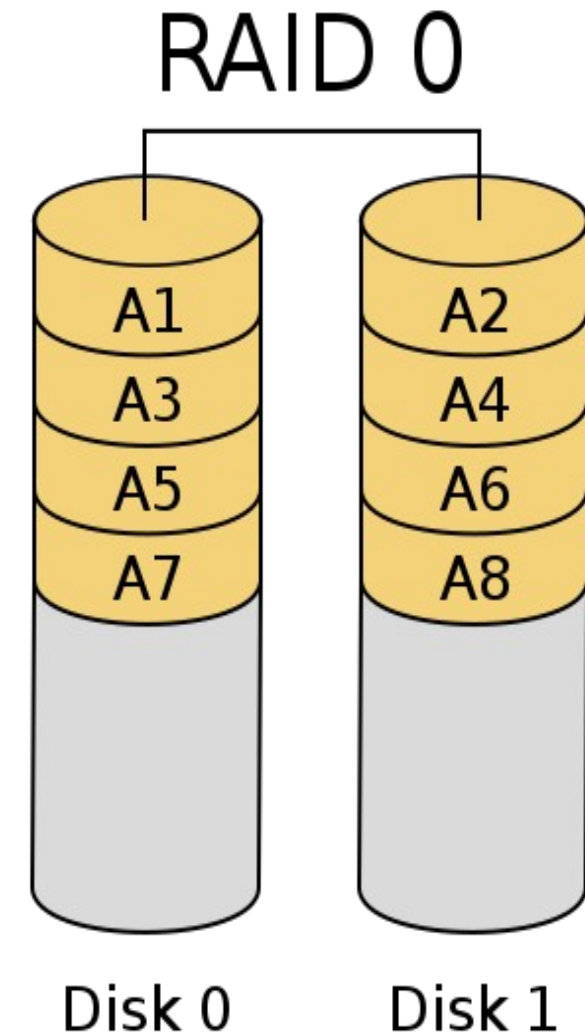
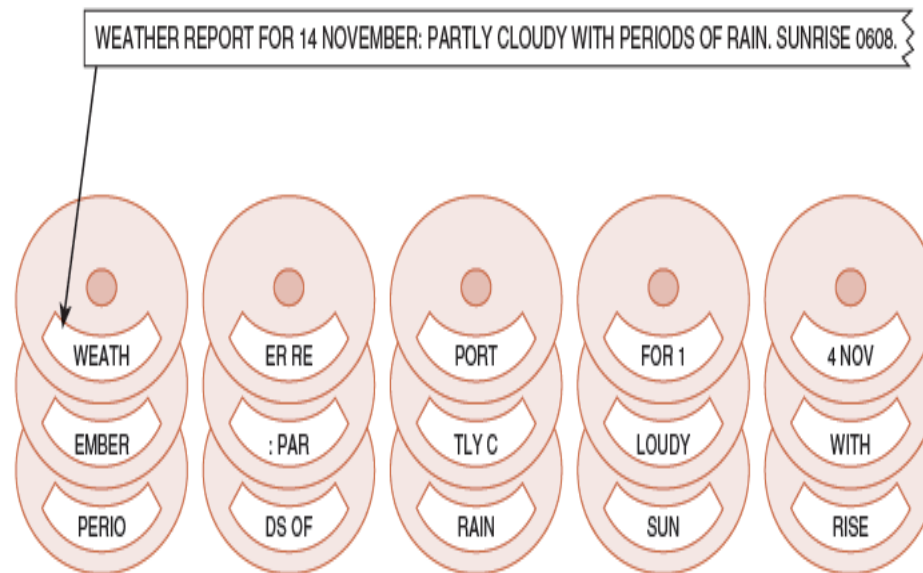


RAID(*Redundant Array of Independent Disks*)

- **RAID**, an acronym **for *Redundant Array of Independent Disks*** was invented to address problems of disk reliability, cost, and performance
- In RAID, data is stored across many disks, with extra disks added to the array to provide error correction (redundancy).

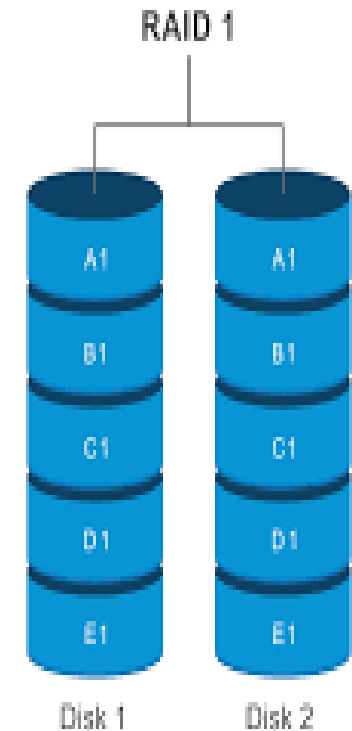
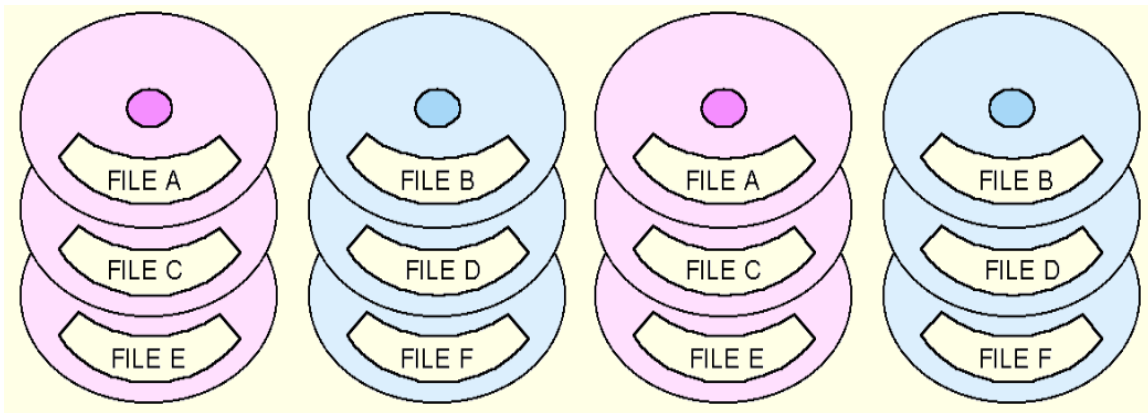
RAID Level 0

- RAID Level 0, also known as ***drive spanning***, provides improved performance, but no redundancy.
 - Data is written in blocks across the entire array.
 - The disadvantage of RAID 0 is in its low reliability.



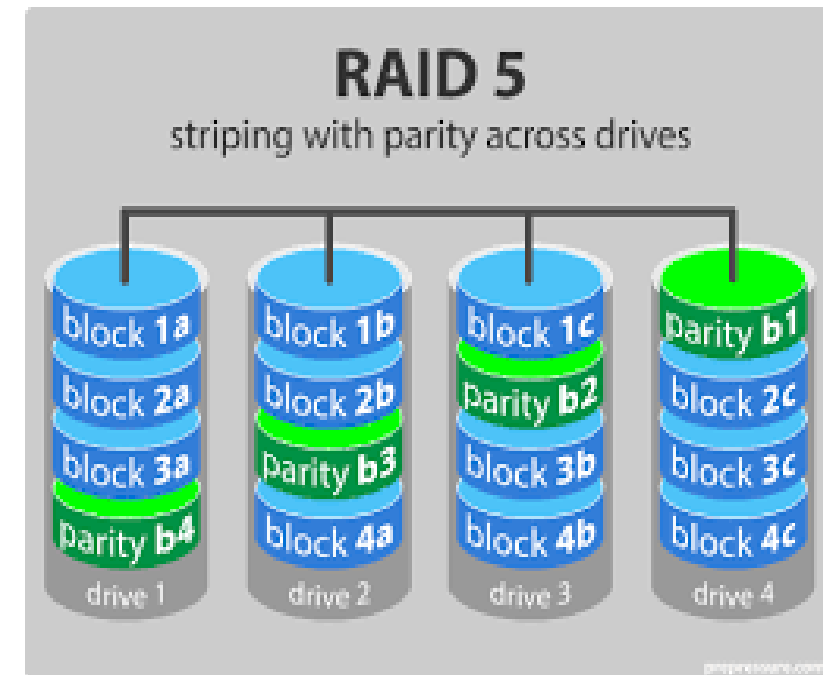
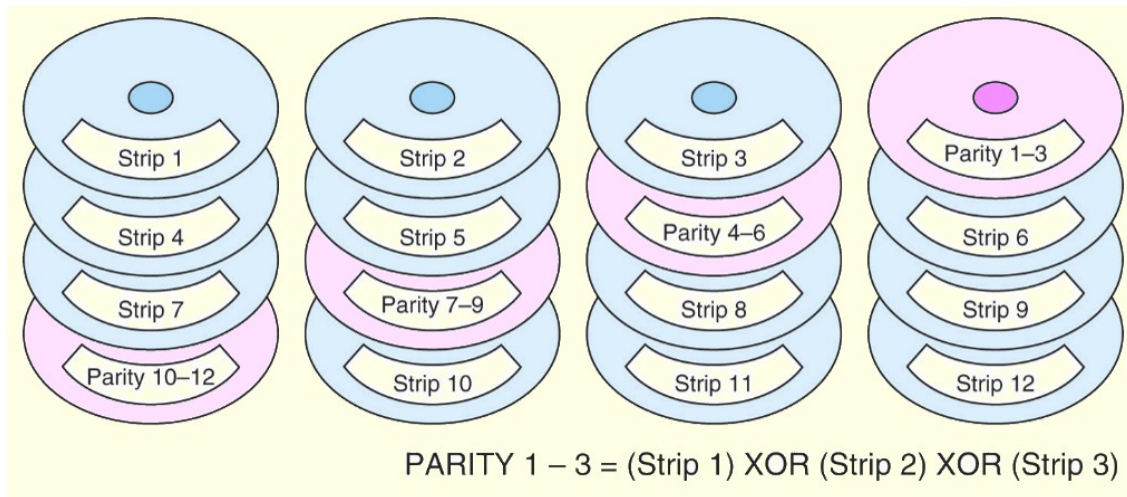
RAID Level 1

- RAID Level 1, also known as ***disk mirroring***, provides 100% redundancy, and good performance.
 - Two matched sets of disks contain the same data.
 - The disadvantage of RAID 1 is cost.



RAID Level 5

- RAID Level 5 is RAID 4 with distributed parity.
 - With distributed parity, some accesses can be serviced concurrently, giving good performance and high reliability.
- RAID 5 is used in many commercial systems.



RAID Level 10

- **RAID 10**, also known as RAID 1+0, is a RAID configuration that combines disk mirroring and disk striping to protect data
- Critical, high-throughput files can benefit from combining RAID 0 with RAID 1, called RAID 10.

